PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · BIOLOGY · 4BI1/1B

A* Examination Guide

Built from 17 mark schemes and 16 question papers, January 2019 to June 2026

This guide is built from the Paper 1B question papers and mark schemes. Model answers follow the marking points examiners list, and every warning comes from an “ignore”, “reject”, “allow” or “no credit” note. Paper 1 tests the core (non-bold) content of the whole specification; it mixes recall, data analysis, calculations and investigation design.

1 How Paper 1B is built
2 Twenty mark-scheme rules
3 Variety of organisms and cells
4 Movement, molecules and enzymes
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9 Coordination and homeostasis
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13 Investigations, graphs and calculations
14 Final A* checklist

Papers analysed: Jan 2019 (legacy 4BI0, mark scheme only), Jun 2019, Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 How Paper 1B is built

Format: 2 hours, 110 marks, 9–11 structured questions: about 1 minute per mark. Every paper has multiple-choice items (with labelled diagrams of cells, flowers or the gut), at least one graph to plot (5–6 marks), several calculations, a 6-mark investigation design and 4–6 mark data evaluations.

Topic	Papers (of 16)	What is asked most
Investigations (CORMS method, 6 marks)	16	design an experiment; control variables; repeats
Graph plotting and data analysis	16	plot (S L A P K), describe trends, evaluate claims
Calculations	16	percentage change, magnification, rate, ratio, standard form, SA:V
Ecology and human impact	14	food webs, pyramids, cycles, quadrats, pesticides, greenhouse gases
Enzymes and respiration	13	temperature effects, yeast, respirometers, limewater/indicator
Plant nutrition and transport	12	starch test, limiting factors, mineral ions, osmosis in potato
Human nutrition and digestion	11	food tests, nutrients and deficiency, gut labels, peristalsis
Reproduction and inheritance	12	flower parts, asexual vs sexual, genetic crosses, pedigrees
Organisms and cells	11	five kingdoms, virus vs cell, organelles, magnification
Coordination, homeostasis, breathing	10	nerves vs hormones, eye, ventilation mechanism
Biotechnology and disease	9	GM, yoghurt, fermenters, antibodies, biological control

2 TWENTY MARK-SCHEME RULES

1. **Never write "amount":** "ignore amount" appears in most control-variable notes. Say volume, mass, concentration, number or type.
2. **Investigation answers follow CORMS:** Change (independent variable), Organisms (same species/age/size), Repeat, Measure 1 (what and how), Measure 2 (after a stated time), Same (two named control variables). One mark each.
3. **Graphs (S L A P K):** linear scale using at least half the grid; straight lines joining the points (or bars, as asked); axes the right way round with labels and units; points within half a small square; a key for two lines. Extrapolating to zero or a bar chart loses L.
4. **Calculations:** correct answer with no working scores full marks, but working earns partial credit; round as asked.
5. **Enzymes are denatured** (active site changes shape, substrate no longer fits); they are not "killed".
6. **Osmosis needs water potential:** water moves from high to low water potential through a partially permeable membrane.
7. **Kinetic energy explanations:** molecules move faster, more collisions, more enzyme–substrate complexes; "faster collisions" is ignored.
8. **Respiration words:** "breathing" or "inhaling" is ignored where respiration is meant; energy is released (ATP), not made.
9. **Word equations score 0 when the balanced symbol equation is asked:** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$.
10. **Comparisons need comparatives** (more, fewer, faster) and both items; quoting figures without a comparator scores 0.
11. **Data evaluations:** for 5–6 marks give points both for and against (sample size, repeats, other variables, species studied, time span, correlation is not causation).
12. **Mineral ions:** name the ion and a matching function (nitrate → amino acids/proteins; magnesium → chlorophyll); "for growth" alone scores nothing.
13. **Tick boxes and MCQs:** one answer per item; two letters or two ticks score 0.
14. **Names must be precise:** gall bladder is rejected for bladder; large intestine for ileum; "intestine" alone may be allowed but not the wrong one.
15. **Pyramids and chains:** a food chain is not a pyramid ("no credit for pyramid"); arrows point to the eater.
16. **Disease:** name the pathogen type and the effect; "harmful" unqualified is ignored.
17. **Pesticide arguments:** bioaccumulation, resistance and non-specific killing; "pollution" unqualified is ignored.
18. **Climate:** link greenhouse gas → heat trapped / less radiation lost → global warming; "climate change" alone is ignored.
19. **Genetics:** genotypes, gametes, offspring, phenotypes, and probability as a fraction, decimal or percentage.
20. **Count the marks and the stem's command:** "describe" = what happens; "explain" = why; a 3-mark explain needs three linked points.

3 VARIETY OF ORGANISMS AND CELLS

Group	Key features	Examples
plants	multicellular, chloroplasts, cellulose cell walls, store starch/sucrose	maize, peas
animals	multicellular, no cell wall, store glycogen, nervous coordination	humans, housefly
fungi	hyphae (mycelium), chitin walls, saprotrophic, extracellular enzymes	Mucor, yeast
bacteria	no nucleus (loop of DNA), plasmids, cell wall, flagellum	Lactobacillus, Pneumococcus
protoctists	mostly single-celled with a nucleus	Amoeba, Chlorella, Plasmodium
viruses	not living cells; protein coat around DNA or RNA; reproduce only in host cells	TMV, flu, HIV

- Characteristics of life: nutrition, respiration, excretion, response, movement, homeostasis, reproduction, growth.
- Virus vs cell (Jun 23, Nov 25): virus smaller, protein coat, no cytoplasm, membrane, ribosomes or organelles. Tobacco mosaic virus causes yellow mosaic leaves (less chlorophyll, less photosynthesis).
- Organelles: nucleus (genetic material), mitochondria (aerobic respiration, ATP), ribosomes (protein synthesis), chloroplasts (photosynthesis; many in palisade cells), vacuole (cell sap), cell wall (strength).
- Tissue: group of similar cells with the same function. Organelle: part of a cell with a particular function.
- Magnification (Nov 25): 43 mm = 43 000 µm; actual = 43 000 ÷ 65 = 660 µm. (Jun 26): 61 000 ÷ 15 000 = 4.1 µm.

4 MOVEMENT, MOLECULES AND ENZYMES

DEFINITIONS

Diffusion: net movement of particles from a region of higher concentration to lower concentration (down a gradient).

Osmosis: water moves from higher to lower water potential through a partially permeable membrane.

Active transport: against a concentration gradient using energy from respiration (ATP).

Enzyme: biological catalyst that speeds up reactions; **metabolism:** all the chemical reactions in an organism.

MODEL ANSWER · POTATO OSMOSIS (5) · NOV 23

Mass is lost in high sucrose concentrations and gained in low concentrations, because water leaves or enters by osmosis from higher to lower water potential; where there is no change in mass the potato and solution have the same water potential. Blot the cylinders dry so surface liquid does not add to the mass.

Percentage change: (2.3 − 2.2) ÷ 2.2 × 100 = 4.5%.

MODEL ANSWER · ENZYME ACTIVITY AND TEMPERATURE (3–4) · JUN 22, JUN 24

As temperature rises, molecules gain kinetic energy, collide more often and form more enzyme–substrate complexes. Above the optimum (about 37–40 °C) the enzyme is denatured: the active site changes shape so the substrate no longer fits. Enzyme concentration graph (Jan 20): rate rises with more enzyme until all substrate is in use, then levels off.

- SA:V (Jun 26): 3 cm cube → 54 : 27 = 2 : 1. Smaller cubes have larger SA:V and are penetrated faster (like single-celled organisms).
- Diffusion rate (Nov 25): 20 cm ÷ 30 s = 0.67 cm/s; higher concentration gives a steeper gradient and faster change.

5 PLANT NUTRITION

PHOTOSYNTHESIS

carbon dioxide + water → glucose + oxygen (light, chlorophyll); $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Light energy is transferred to chemical energy.

Limiting factors: light intensity, CO_2 concentration, temperature. Where a curve levels off, another factor is limiting.

STARCH TEST ON A LEAF (JUN 25, NOV 25)

Destarch the plant by keeping it in the dark for 24–48 hours (starch is used in respiration); after the experiment boil the leaf in water, then in ethanol (water bath, no flame) to remove chlorophyll, rinse, add iodine: blue-black = starch.

Controls: foil or stencil for light; soda lime (no CO_2) vs none; variegated leaves for chlorophyll.

- Pondweed (Jun 21, Jun 26): count bubbles per minute; change light distance or bulb; keep temperature, CO_2 and plant the same.
- Mineral ions: nitrate → amino acids/proteins; magnesium → chlorophyll; phosphate → DNA/ATP; potassium → enzymes.
- Waterlogged soil (Jun 25): little oxygen, so little root respiration, so little active transport of mineral ions.
- Leaf adaptations: large surface area, thin, palisade with many chloroplasts, stomata, air spaces.

6 HUMAN NUTRITION

Test	Method	Positive
starch	iodine solution	orange-brown → blue-black
glucose (reducing sugar)	Benedict's, heat in a water bath (~70 °C)	blue → green/yellow/orange/brick red
protein	biuret	blue → purple
lipid	crush, ethanol, then water (emulsion test)	milky white emulsion

- Nutrients: carbohydrate (energy), protein (growth, enzymes), lipid (energy store, insulation), fibre (peristalsis, prevents constipation), vitamin A (vision), C (prevents scurvy), D (bone growth, calcium absorption), calcium (bones/teeth), iron (haemoglobin).
- Energy needs (Jun 26): rise with body mass and activity; protein for growth; iron rises at menstruation and pregnancy.
- Peristalsis: circular and longitudinal muscles contract behind the bolus and push it along.
- Bile: emulsifies fat (larger surface area for lipase) and neutralises acid. Villi increase surface area for absorption.
- Diet evaluations (Jan 23, Nov 24): compare each nutrient with its use; too much carbohydrate → fat → obesity, diabetes, CHD.

7 RESPIRATION AND GAS EXCHANGE

RESPIRATION

Aerobic: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$, releasing energy (ATP). Anaerobic in muscle: glucose → lactic acid; in yeast: glucose → ethanol + carbon dioxide.

Respirometer (Nov 20): KOH absorbs CO_2; the liquid moves towards the organisms as oxygen is used. Control variables: mass of organisms, temperature, volume of KOH, time.

Limewater or hydrogencarbonate indicator: exhaled air turns limewater cloudy faster (more CO_2 from respiration).

MODEL ANSWER · INHALATION (4–6) · JAN 22, NOV 24

Diaphragm contracts and flattens; external intercostal muscles contract; ribcage moves up and out; volume of the thorax increases; pressure decreases; air is drawn into the lungs. Label order: trachea, bronchi, bronchioles, alveoli.

- Lung damage (dust, smoking, emphysema): less surface area, longer diffusion distance, less oxygen into the blood, less aerobic respiration, more anaerobic respiration and lactic acid.
- Exercise: breathing rate rises for oxygen and to remove CO_2; stays high afterwards to remove lactic acid (oxygen debt).
- Germinating seeds produce heat (thermometer); boiled seeds act as a control.

8 TRANSPORT IN PLANTS AND ANIMALS

- Plasma carries glucose (respiration), amino acids (protein synthesis), iron (haemoglobin), water, CO_2, urea, hormones.
- Red blood cells: haemoglobin, no nucleus, biconcave; white cells: phagocytes engulf, lymphocytes make antibodies.
- Antibodies (Jan 23): specific to an antigen; they clump or label pathogens so they are destroyed.
- Vectors (Jun 24): a tick sucks blood from an infected animal and passes the pathogen to the next animal it bites.
- Coronary heart disease risk factors: genetics, high blood pressure, high-fat diet, smoking, stress, lack of exercise, obesity.
- Plants: xylem carries water and minerals; phloem carries sucrose and amino acids; root hair cells have a large surface area.

9 COORDINATION AND HOMEOSTASIS

	Nervous	**Hormonal**
signal	electrical impulses along neurones	chemicals in the blood plasma
target	specific	all around the body (target organs)
speed and duration	fast, short-lived	slower, longer-lasting

- Speed ratio (Jun 22): nerve 55 m/s vs hormone 4.2 m/min = 0.07 m/s ⇒ 786 : 1.
- Eye: retina receptors send impulses along the optic nerve; damage → loss of vision.
- Plant tropisms (Jan 21): roots grow towards gravity (positively gravitropic) and away from light; shoots the converse; auxin.
- Hormones: insulin and glucagon (pancreas), adrenaline, oestrogen and progesterone (ovary), testosterone (testis), ADH.

- Puberty (Jun 21): testosterone in males (muscle, deeper voice, facial hair); oestrogen in females (breasts, hips, menstruation).

10 REPRODUCTION AND INHERITANCE

- Flower parts: anther (pollen), filament, stigma (pollen lands and germinates), style, ovary (becomes fruit), ovule (becomes seed), petals. Wind-pollinated flowers: exposed anthers and feathery stigmas, small green unscented petals.
- Asexual (runners, cuttings, tubers, tissue culture): clones, no variation, fast, keeps desired characteristics. Sexual: variation, new varieties by cross-pollination and selective breeding (Nov 24).
- Germination needs water, oxygen, a suitable temperature. Salt lowers soil water potential, so less water enters by osmosis (Jan 21).
- Fertilisation: fusion of male and female gamete nuclei ("meets" is ignored). Human gametes have 23 chromosomes.
- Sperm: nucleus with chromosomes, acrosome, many mitochondria for respiration to swim.
- Crosses: Dd × dd gives 1 : 1; Aa × Aa gives 3 : 1. Sickle cell: recessive, only expressed in the homozygote (Jan 21); for a child of a stated sex, multiply by ½ (affected boy from two carriers = ¼ × ½ = 1/8).
- Natural selection (Jan 19, Jan 23 moths): variation by mutation; better camouflaged individuals survive predation, reproduce and pass on the allele; percentage change 75% → 41% (34 percentage points).

11 ECOLOGY AND HUMAN IMPACT

- Food chains start with a producer; arrows show energy flow. Pyramid of numbers or biomass must be drawn to scale.
- Energy loss between levels: respiration (heat), movement, excretion, egestion (undigested cellulose in faeces), uneaten parts.
- Decomposers (bacteria, fungi): saprotrophic, release enzymes, digest dead matter, respire releasing CO_2.
- Carbon cycle: photosynthesis removes CO_2; respiration, combustion and decomposition return it.
- Quadrats (Jun 21): place randomly (coordinates), count, repeat, mean per quadrat × total area.
- Population changes (Nov 23): removing a predator or prey changes food supply; consider disease, migration, pesticide.
- Biological control (Jun 21, Nov 23): introduce a predator (ladybirds eat greenfly); specific, no reapplication, no bioaccumulation, no resistance; but could become a pest itself.
- Greenhouse gases: CO_2 from fossil fuels; methane from cattle and paddy fields; trap heat → global warming; melting ice affects food chains (Nov 20).

12 MICROORGANISMS, BIOTECHNOLOGY, DISEASE

- Yoghurt: pasteurise the milk to kill other bacteria (prevents competition), add Lactobacillus, incubate; lactose → lactic acid.
- Yeast: anaerobic respiration produces ethanol and CO_2; growth stops as glucose runs out or ethanol builds up (Jun 21).
- Genetic engineering (Nov 20): restriction enzyme cuts the plasmid and the gene (sticky ends); ligase joins them; vector (plasmid or virus) carries the gene into the organism.
- GM crops debate (6): fewer pesticides, specific to pests; but cross-pollination, harm to pollinators, resistance, consumer concern, seed patents.
- Disease examples: Plasmodium → malaria; Pneumococcus → pneumonia; TMV → mosaic leaves. Immunity from antibodies.

- Drug and treatment trials (Nov 21 statins): consider sample size, duration, chance, pre-existing conditions, lifestyle differences.

13 INVESTIGATIONS, GRAPHS AND CALCULATIONS

CORMS TEMPLATE (6 MARKS)

C change: use at least five values of the independent variable. **O** organisms: same species, age, size. **R** repeat at each value and calculate a mean. **M1** measure the dependent variable with named apparatus. **M2** after a stated time. **S** keep two named variables the same (temperature, volume, concentration, light).

Calculation	Past-paper example
percentage change	(0.08 − 0.04) ÷ 0.04 × 100 = 100% (Jun 19); (2.2 − 1.25) ÷ 1.25 × 100 = 76% (Jun 21)
percentage of a total	356 ÷ 1687 × 100 = 21.1% (Jan 22); 110 ÷ 453 × 100 = 24% (Jun 24)
ratio	85 147 : 7206 = 11.8 : 1 (Jan 22); 786 : 1 (Jun 22)
rate	6.0 ÷ 30 = 0.20 (Nov 21); 530 ÷ 30 = 17.7 (Jan 20)
area/volume	$\pi \times 0.05^2 \times 0.6 = 4.7 \times 10^{-3}$ (Jun 19); 0.55 × 12.56 = 6.9 mm^2 (Nov 21)
standard form	$2.5 \div 15\,000 = 1.7 \times 10^{-4}$ (Jun 21); 24.8% of 58 million = 1.4×10^7 (Jun 19)
time conversion	10 000 ÷ 1.6 = 6250 s ÷ 60 = 104 min (Nov 20)

- Describing a graph: trend + numbers with units + changes in gradient + levelling off; then explain with biology.
- Median (Jun 22) is better than the mean when data are skewed or have extreme values.
- Reliability: repeat and calculate a mean, identify anomalies; validity: control the other variables.

14 FINAL A* CHECKLIST

1. Investigations: CORMS, named apparatus, never “amount”.
2. Graphs: S, L, A, P, K — five or six marks every paper.
3. Calculations: show steps, convert units (mm → µm × 1000), round as asked.
4. Enzymes: kinetic energy, collisions, enzyme–substrate complexes, denatured active site.
5. Osmosis: water potential and a partially permeable membrane.
6. Equations: balanced symbol equations for respiration and photosynthesis.
7. Evaluations: points for and against; sample size, repeats, other variables.
8. Genetics: full diagram, probability with sex if asked. Leave 10 minutes to check graphs and calculations.